

Report 1: Initial Kagome-VQE Project Note

Adel H. Al-Yoorby
QIntern 2026 Project qi26_09
Project context: Dr. Muhammad Ahsan

July 2026

Potential solution. This project explores a no-calibration route for preparing low-energy states of the 19-site Kagome antiferromagnetic Heisenberg benchmark. Instead of changing the Hamiltonian, the idea is to start from a physically motivated signed weighted-RVB state and then refine it with shallow edge-colored Heisenberg-HVA circuit layers. The exact benchmark is

$$E_0 = -29.146168109350,$$

while the current preliminary no-calibration result reaches

$$E = -29.037601351802, \quad F = 0.982518440088$$

at HVA depth $p = 4$. This is not a final proof of scalable quantum spin liquid preparation; it is an initial finite-size proof-of-concept showing that better state structure plus Heisenberg-level circuit refinement can move a dimer/RVB state close to the exact 19-site ground state.

Sketch.



Current interpretation. The weighted-RVB initializer captures most of the useful finite-size state structure, and the Heisenberg-HVA layers give a smaller but real circuit-compatible refinement. Ahsan-style calibrated-Hamiltonian scans are useful as methodology context and as clues for future ansatz design, but the main current direction is to keep the target Hamiltonian unchanged and improve the circuit/state ansatz.

Repository / current work link. <https://github.com/adoolelomani2026/qintern-kagome-hva>

Next step. The next controlled experiment is to test calibration-informed local circuit blocks, such as adding extra Heisenberg gates on the important triangle (6, 7, 8) or bond (14, 16), while still evaluating against the original unmodified Hamiltonian.

